KNN

Classification (predicting classes)

e.g Predicting moving Genre

IMDb Rating	Duration	Genre
8.0 (mission impossible)	160	Action
6.2 (Gadar 2)	170	Action
7.2 (Rocky and Rani)	168	Comedy
8.2 (OMG2)	155	Comedy

Now predict the genre of "Barbie" move with IMDb rating 7.4 and duration 114

Step1: Calcullation Distance (Euclidean distance)

Calculate the Euclidean distance b/w the new movie and each movie in the dataset

$$d(p,q) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$x_1 = 7.4$

$y_1 = 114$

$x_2 \quad y_2$

Distance to $(8.0, 160) = \sqrt{((7.4 - 8.0)^2 + (114 - 160)^2)} = \approx 46.00$

Distance to $(6.2, 160) = \sqrt{((7.4 - 6.2)^2 + (114 - 170)^2)} = \approx 56.01$

Distance to $(7.2, 168) = \sqrt{((7.4 - 7.2)^2 + (114 - 168)^2)} = \approx 54.00$

Distance to $(8.2, 155) = \sqrt{((7.4 - 8.2)^2 + (114 - 155)^2} = \approx 41.00$

Step2:

Select k Nearest Neighbors

Suppose $k = 1$ (Neighbors)

Small distane 41.00

Best value 5

$k = 3$

Step3: Majority voting (A, C, C)

classification

Barbie move "C"

K-mean clustering algorithm

Sr. No	age	amount
C1	20	500
C2	40	1000
C3	30	800
C4	18	300
C5	28	1200
C6	35	1400
C7	45	1800

Centroid value

Step 1: Suppose two cluster

K1 K2

Step2: Centroid value (Random Select)

K1 K2

C1 20,500

C2 40,1000

Step3: C3 = Nearest distance check it

By using Euclidean Distance

obj Centroid

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Step 4:

$$K1 \rightarrow C3 = \sqrt{(30-20)^2 + (800-500)^2}$$

$$C3 = \sqrt{(10)^2 + (300)^2} = \boxed{300.0}$$

$$K2 \rightarrow C3 = \sqrt{(30-40)^2 + (800-1000)^2} = \boxed{200.0}$$

distance small

K1	K2
C1	C2, C3
20, 500	40, 1000
	30, 800

After assign the value must update the Centroid value.

$$\frac{800+1000}{2} = \boxed{900} \; ; \; \frac{40+30}{2} = \boxed{35}$$

Step 5:

K1

C1

20,500

K2

C2, C3

35,900

Step 6:

$$C_4 = \sqrt{(18-20)^2 + (300-500)^2} = \text{small vale}$$

$$C_4 = \sqrt{(18-35)^2 + (300-900)^2} =$$

K1

C1, C4

~~20,500~~

29, 650

K2

C2, C3

35, 900

$$\frac{500+300}{2} = 650$$

$$\frac{20+18}{2} = 29$$

.... upto C7